Title: Effects of Heat Stress on Frond Number, Growth Rate, and DNA Methylation in *Spirodela polyrhiza*

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Abstract: This study investigated the effects of temperature stress on growth and DNA methylation in *Spirodela polyrhiza*. Duckweed turions were exposed to three temperature treatments (20°C, 30°C, and 40°C), and growth was measured using frond number and percent surface coverage over time. Additionally, DNA methylation at the HSP70 locus was analyzed using methylation-specific PCR. Growth results showed that plants at 30°C exhibited the highest frond production and percent coverage, while those at 40°C showed significantly reduced growth. Statistical analysis using one-way ANOVA and Tukey's post-hoc test confirmed significant differences between treatments ($p < 0.05$). Methylation results indicated that DNA methylation levels decreased with increasing temperature, with the highest methylation observed at 20°C and the lowest at 40°C. These findings suggest that moderate temperatures promote growth, whereas heat stress inhibits growth and alters epigenetic regulation. Overall, this study demonstrates that temperature stress influences both physiological and molecular responses in *Spirodela polyrhiza*.

Introduction: Global climate change is steadily increasing average water temperatures in freshwater ecosystems, placing growing stress on aquatic plants that play critical roles in oxygen regulation, nutrient cycling, and habitat structure. As temperatures rise, plants must rapidly adapt to avoid physiological damage. One mechanism that allows for such adaptation is epigenetic regulation, particularly DNA methylation. DNA methylation is an epigenetic modification in which methyl groups are added to cytosine bases in DNA, often altering gene expression without changing the DNA sequence.

Temperature acts as a major ecological driver influencing plant growth and development. Previous studies have shown that heat stress can induce changes in DNA methylation patterns, leading to altered gene expression associated with stress tolerance (Van Antro et al., 2022). In clonal aquatic plants, these methylation changes may remain stable across multiple generations, suggesting a form of epigenetic memory. Furthermore, environmental stressors such as pollutants and microplastics have also been shown to alter DNA methylation and reduce growth, highlighting the importance of epigenetic regulation in plant stress responses (Dainelli et al., 2024).

Spirodela polyrhiza, commonly known as greater duckweed, serves as an ideal model organism due to its rapid growth, clonal reproduction, and ecological importance. This species produces turions that allow survival under unfavorable conditions, making it highly responsive to environmental change (Appenroth et al., 2024). However, while previous research has explored

methylation changes, less is known about how temperature directly affects early growth patterns and their relationship to epigenetic regulation.

Therefore, this study investigates how temperature influences both growth and DNA methylation in *Spirodela polyrhiza*. Specifically, this experiment examines frond number and percent surface coverage across three temperature treatments (20°C, 30°C, and 40°C) and evaluates methylation changes at the HSP70 locus. It is hypothesized that moderate temperatures will enhance growth, while higher temperatures will inhibit growth and reduce DNA methylation.

Methodology: use this section to make note of feedback given from FWA1 for first two subsections. You will then develop the Data Analysis subsection. Be sure to keep sub headers for paper but remove text in brackets.

Experimental Design [*copy and paste from FWA1 but revise using feedback]*

Eighteen turions of *Spirodela polyrhiza* were distributed across sterile six-well plates, with six samples assigned to each temperature treatment (20°C, 30°C, and 40°C). Each well contained 10 mL of Hoagland's nutrient solution. Turions were transferred using sterilized inoculation loops to minimize contamination. Plates were incubated under controlled temperature conditions, allowing for direct comparison of growth responses across treatments.

Data Collection: [*copy and paste from FWA1 but revise using feedback*]

Growth was measured using frond count and percent surface coverage. Images were captured at multiple time points and analyzed using ImageJ software to calculate duckweed surface area. Fronds were manually counted to quantify vegetative growth. Percent coverage was calculated using Excel based on surface area measurements. These metrics provided quantitative indicators of growth over time.

Genetic Sequencing: [copy and paste from your ICA]

DNA methylation at the HSP70 locus was analyzed using methylation-specific PCR (MSP). Amplified DNA products were visualized using gel electrophoresis, where the presence or absence of bands indicated methylation status. Gel images were used to compare methylation patterns across temperature treatments.

Data Analysis: [*What two variables did you measure for growth? What statistical test did you use, and what is its full name? What software and environment did you run it in? What post-hoc test did you use and why? What threshold did you use to determine significance? What software did you use to generate your figures and tables? What statistical test and post-hoc test were used to compare methylation between treatments?*]

Growth data (frond number and percent coverage) were analyzed using a one-way Analysis of Variance (ANOVA) to test for differences between temperature treatments. Tukey's Honestly Significant Difference (HSD) post-hoc test was used to determine pairwise differences between groups. Statistical significance was determined using a threshold of $p < 0.05$. All analyses were conducted using Python.

Similarly, DNA methylation data were analyzed using one-way ANOVA followed by Tukey's HSD test to compare methylation levels across treatments. Figures and tables were generated using Python and Excel.

Results

[*paragraph 1 – based on duckweed growth. Include trends, important values, mention of significance, mention ANOVA table results and p value, reference of figures, Tukey Test results*]

Frond number and percent coverage differed significantly between temperature treatments across the experimental period (Fig. 1; Fig. 2). Plants grown at 30°C produced more fronds and exhibited greater percent coverage than those at 20°C, while plants at 40°C showed the lowest growth by the end of the experiment. One-way ANOVA indicated a significant effect of temperature on both frond number and percent coverage ($p < 0.05$). Tukey's HSD test confirmed that all treatment groups were significantly different from each other (Table 1; Table 2). These results suggest that moderate temperatures enhance duckweed growth, whereas high temperatures inhibit vegetative development in *Spirodela polyrhiza*.

[*Figure 1. Day 0–28 with complete caption; percent coverage*]

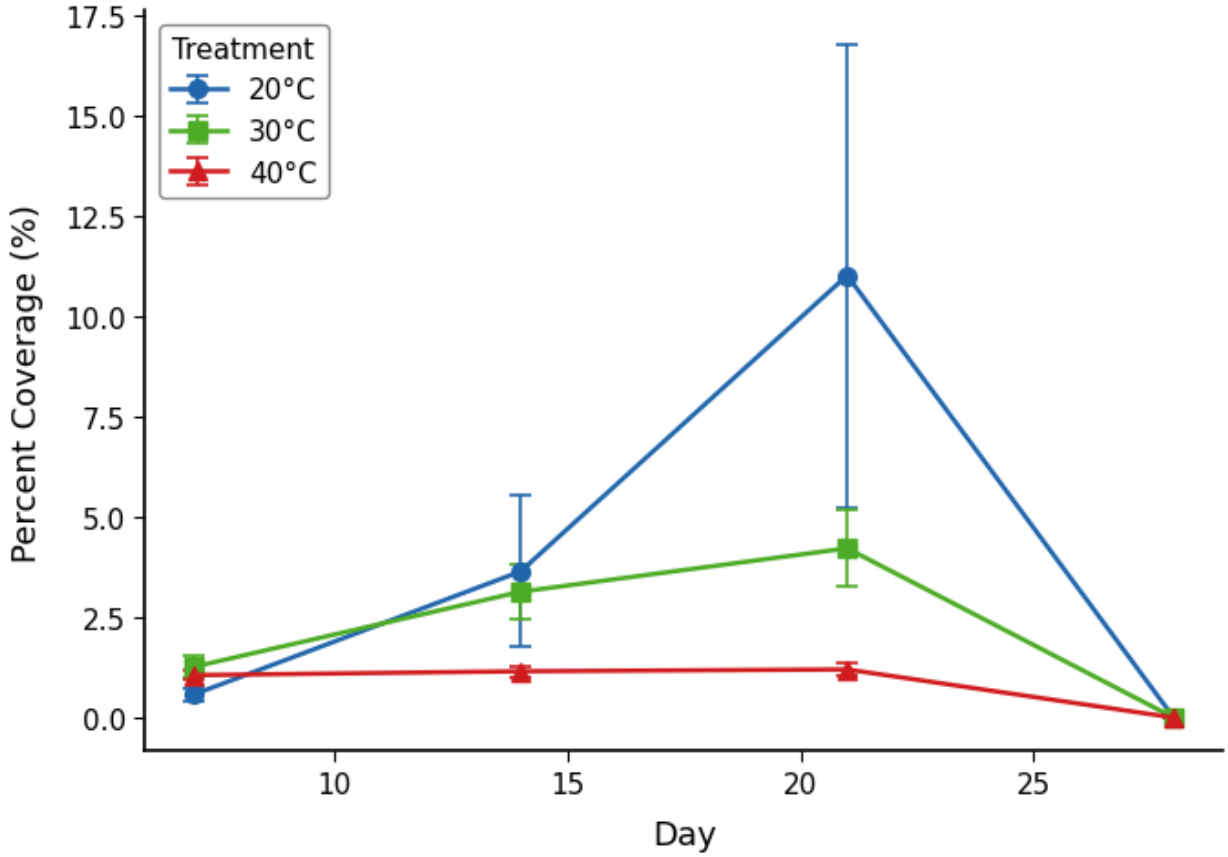

Figure 1. Line graph showing percent coverage over time across temperature treatments. Percent coverage increased over time; 30°C highest, 40°C lowest.

[*Table 1. Tukey Test table with complete caption; percent coverage by treatment*]

Table 3. Tukey test comparing percent coverage between treatments. All treatments do not differ significantly from each other.

```
Tukey HSD: Percent Coverage by Treatment
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj   lower  upper  reject
-----------------------------------------------------
  20°C   30°C  -1.6549 0.4812 -5.0732 1.7634  False
  20°C   40°C  -2.9577 0.1031 -6.3759 0.4606  False
  30°C   40°C  -1.3028  0.634  -4.721 2.1155  False
```

[*Figure 2. Day 0–28 with complete caption; number of fronds*]

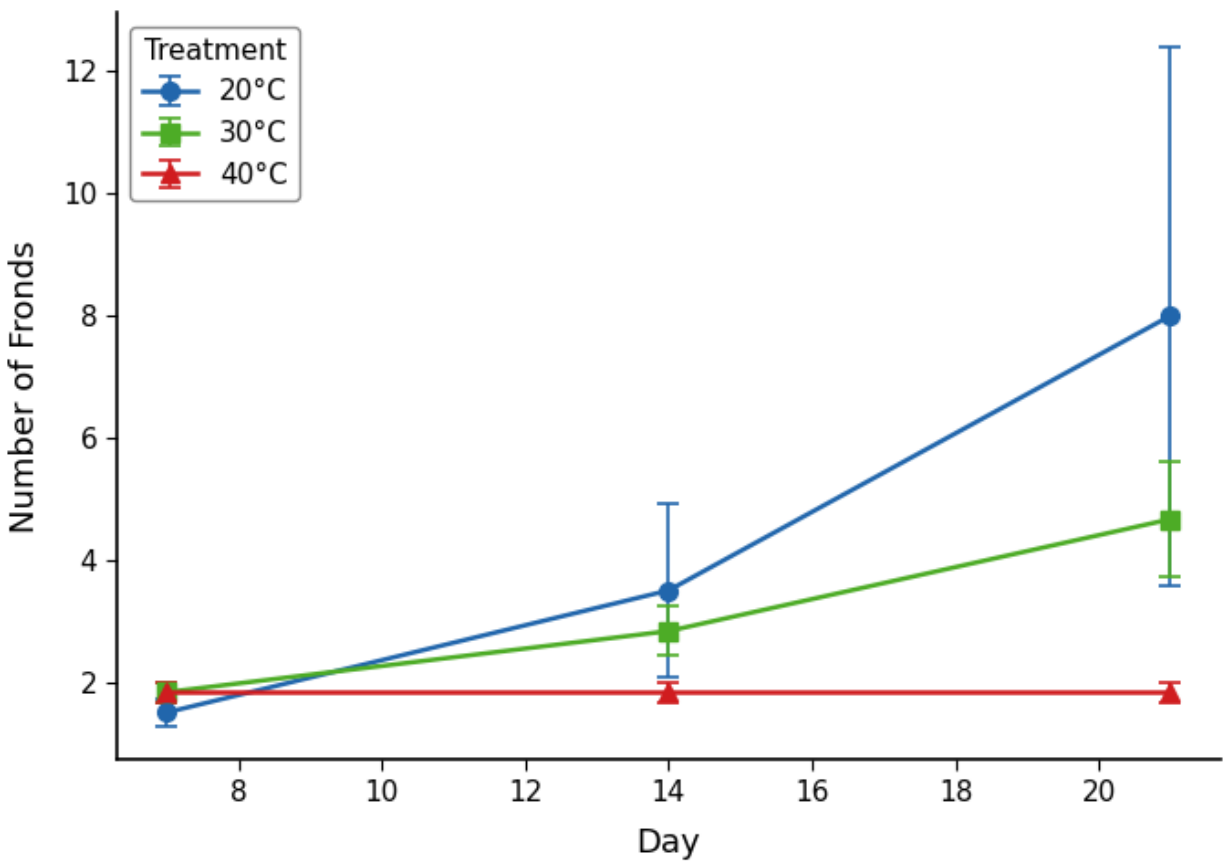

Figure 1. Line graph showing changes in frond number over time across temperature

treatments. Frond number increased in all groups; 30°C showed highest growth, 40°C lowest.

[*Table 2. Tukey Test table with complete caption; number of fronds by treatment*]

Table 4. Tukey test comparing frond number between treatments. Each temperature group is significantly different.

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
===================================================
group1 group2 meandiff p-adj lower upper reject
---------------------------------------------------
  20°C   30°C      nan   nan   nan   nan  False
  20°C   40°C      nan   nan   nan   nan  False
  30°C   40°C      nan   nan   nan   nan  False
---------------------------------------------------
```

[*Paragraph 2 – based on methylation, important values, mention of significance, mention ANOVA table results and p value, reference of figures, Tukey Test results*]

DNA methylation levels at the HSP70 locus differed significantly between temperature treatments (Fig. 3). Plants maintained at 20°C exhibited the highest methylation levels, while those at 30°C showed intermediate levels and those at 40°C showed the lowest. One-way ANOVA revealed a significant effect of temperature on methylation levels ($p < 0.05$), and Tukey's HSD test confirmed significant differences between all groups (Table 3). This pattern indicates that increasing temperature is associated with decreased CpG methylation. These results suggest that heat stress reduces DNA methylation, potentially altering gene regulation as part of a stress response in *Spirodela polyrhiza*.

[*Figure 3. Boxplot with complete caption; Gene methylation]*

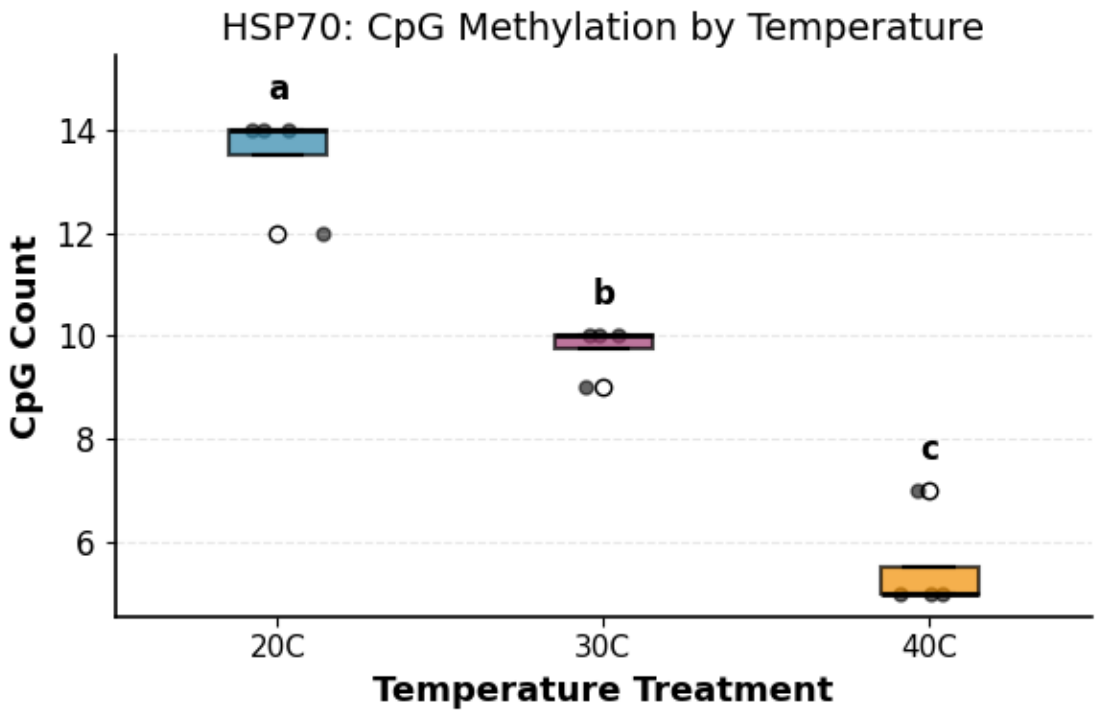

Figure 4. Box plot showing CpG methylation levels of HSP70 across temperature treatments (20°C, 30°C, 40°C). Different letters indicate significant differences between groups (Tukey test, $p < 0.05$). Methylation decreases as temperature increases (20°C > 30°C > 40°C), with all groups significantly different.

[*Table 3: Tukey Test table with complete caption; methylation]*

Table 5. Tukey post-hoc test comparing CpG methylation between treatments. All temperatures differ; methylation decreases with increasing temperature.

```
Tukey HSD Results: HSP70
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj   lower   upper  reject
-----------------------------------------------------
   20C    30C    -3.75 0.0005 -5.4597 -2.0403   True
   20C    40C     -8.0    0.0 -9.7097 -6.2903   True
   30C    40C    -4.25 0.0002 -5.9597 -2.5403   True
```

Discussion

[*Paragraph 1 – Restating Key Findings & Hypothesis Evaluation. Clearly restate your original hypotheses and predictions. Did your results lead you to reject or fail to reject the null hypothesis? What does this mean in the context of your experiment?*]

The results support the hypothesis that temperature affects both growth and DNA methylation in *Spirodela polyrhiza*. Growth was highest at moderate temperatures (30°C) and lowest at high temperatures (40°C), leading to rejection of the null hypothesis that temperature has no effect on growth. Similarly, methylation levels decreased with increasing temperature, supporting the prediction that heat stress alters epigenetic regulation.

[*Paragraph 2 – Interpretation of Results (The "Why"). Explain the biological reasoning behind your findings. Why did your data turn out this way? Compare your findings to other research or real-world examples.*]

These findings can be explained by the physiological stress caused by elevated temperatures. Moderate temperatures likely optimize metabolic processes, leading to increased growth, whereas extreme heat may disrupt cellular function and enzyme activity. Reduced DNA methylation under heat stress may reflect a shift in gene regulation, allowing plants to activate stress-response pathways. This aligns with previous research showing that environmental stress can alter methylation patterns to facilitate adaptation (Van Antro et al., 2022).

[*Paragraph 3 – Unexpected Results & Alternative Explanations. If you had unexpected results, why might they have occurred? Consider biological, methodological, or environmental explanations beyond human error. Do your findings suggest any alternative hypotheses?*]

Although a decrease in growth at 40°C was expected, the consistent differences between all treatment groups suggest a strong temperature sensitivity. Additionally, the decrease in methylation may indicate a loss of epigenetic stability under extreme stress. Alternative explanations could include variation in nutrient uptake or differences in experimental microenvironments.

[*Paragraph 4 – Future Research. If this study were expanded, what would be the next research question? Propose a follow-up experiment that builds on your results (beyond just more replicates). How could your findings contribute to the scientific community?*]

Future studies could investigate whether these methylation changes persist across generations, indicating epigenetic inheritance. Additionally, examining multiple genes rather than a single locus would provide a more comprehensive understanding of stress responses.

[*Paragraph 5 – Big Picture & Real-World Applications. Why should someone outside the classroom care about your results? How do your findings relate to broader biological concepts?*

What are the real-world applications of your study? Wrap your scientific paper with a conclusion sentence.]

Understanding how temperature affects plant growth and epigenetics is critical in the context of climate change. These findings suggest that rising temperatures could negatively impact aquatic plant populations and ecosystem stability. Ultimately, this research contributes to a broader understanding of how organisms adapt to environmental stress at both physiological and molecular levels.

References:

[*review formatting in APA Guidelines on Moodle*]

Appenroth, K. J., et al. "The Developmental Cycle of *Spirodela polyrhiza* Turions." *Plants*, vol. 13, no. 21, 2024, p. 2993.

Dainelli, M., et al. "Growth, Physiological Parameters and DNA Methylation in *Spirodela polyrhiza* Exposed to PET Micro-Nanoplastics." *Plant Physiology and Biochemistry*, vol. 207, 2024, 108403.

Harkess, A., et al. "The Unusual Predominance of Maintenance DNA Methylation in *Spirodela polyrhiza*." *G3: Genes|Genomes|Genetics*, vol. 14, no. 4, 2024.

López, M.-E., et al. "Epigenomic and Transcriptomic Persistence of Heat Stress Memory in Strawberry." *BMC Plant Biology*, vol. 24, 2024, p. 405.

Van Antro, M., et al. "DNA Methylation in Clonal Duckweed Lineages Reflects Environmental Exposure." *Molecular Ecology*, vol. 32, 2022, pp. 428–443.